**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Flowcharts and Algorithm Design

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. For cost = price x quantity, identify input, processing and output.

2. What decides whether an action is legal?

A) Its popularity
B) Its speed
C) Its cost
D) The law

3. What is a privacy concern with collecting personal data?

4. An online service uses only one language. "Technology cannot affect language or who takes part." Correct this. Who might be left out?

## Today's objectives

Recognise the purpose of standard OCR J277 flowchart symbols. Trace a Yes/No decision for two values.

## Key vocabulary - Read, cover, recall

**Flowchart**
Diagram of algorithm steps and control flow.

**Terminal**
Start or End symbol.

**Decision**
Diamond evaluating a condition.

**Process**
Rectangle for a calculation or instruction.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Reading a flowchart** | A flowchart represents an algorithm visually. Follow arrows from Start through each process or decision until End. |
| **Use the right shapes** | Start and End use terminal shapes. Inputs and outputs use parallelograms, processes use rectangles, and decisions use diamonds. Predefined subprograms have their own symbol. |
| **Decisions choose one branch** | A diamond evaluates a condition. True/Yes and False/No lead down alternative routes, so a value follows only the appropriate path. |
| **Correct loops and missing paths** | Follow the algorithm with a test value. If input is invalid, an arrow can return to input/check for another attempt. Each branch should lead to a defined next action or End. |
| **Structure diagrams** | A structure diagram shows decomposition: Quiz has Menu, Play and Results beneath it; Play has Input and Scoring beneath it. Links show the hierarchy between the subproblems. Unlike a flowchart, this diagram does not describe the runtime order of decisions. |

## I Do - One worked example

**A flowchart checks if a mark is at least 5. If true it prints 'Pass'; otherwise it prints 'Try again'. Explain which branch is followed for 7 and 3. [4 marks]**

1. The diamond checks mark >= 5. Only one branch is followed per test.
2. 7 >= 5 is true [1], so the Yes branch prints Pass [1].
3. 3 >= 5 is false [1], so the No branch prints Try again [1].
4. Check both comparisons and the different outputs; avoid running both branches for one input.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A flowchart checks whether a number is greater than 10. Explain the route and output for 12 and 8. [4 marks]

2. A lunch-order flowchart repeats a question after an invalid choice. Explain why an arrow loops back to the input. [4 marks]

3. A flowchart uses a rectangle 'total = a + b' then a parallelogram 'Display total'. Explain both symbols. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Predict whether this image file fits.**

```
freeMB = 100
imageMB = 80
if imageMB <= freeMB then
print("Fits")
else
print("Too large")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State which flowchart shape represents a decision. [1 mark]

Q3. A flowchart checks if a customer is at least 16 years old before printing Eligible. Explain what happens for 18 and 14. [4 marks]

Q4. Describe the purposes of terminal, input/output, process and decision symbols in flowcharts and the use of arrows. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A flowchart's decision diamond can direct execution down one of two branches. ☐ T ☐ F
2. A flowchart represents an algorithm visually ☐ T ☐ F
3. Start and End use terminal shapes ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Inputs: price and quantity. Processing: multiplication. Output: total cost. ☐

**Q2** D - The law ☐

**Q3** People may lose control over how identifying information is used or shared. ☐

**Q4** Technology can change language, access, traditions and who can take part. Consider who benefits or is harmed. Something legal may still be unfair. ☐

## Code First - Model

```
freeMB = 100
imageMB = 80
if imageMB <= freeMB then
print("Fits")
else
print("Too large")
endif
```

**Output: Fits**

## We Do - Check the model

1. 12 > 10 is true [1], follows Yes and prints High [1]. 8 > 10 is false [1], follows No and prints Low [1]. ☐

2. Invalid choice follows the No route [1] and causes another input request [1]. The return arrow rechecks the new choice [1] until it is valid [1]. ☐

3. Rectangle represents processing [1] calculating and storing a sum [1]. Parallelogram represents output [1], displaying the calculated total [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** A diamond [1]. ☐

**Q3** 18 >= 16 is true [1] so Eligible prints [1]. 14 >= 16 is false [1], so the not-eligible alternative prints [1]. ☐

**Q4** Terminal identifies start/end [1]; parallelogram input/output [1]; rectangle performs process/calculation [1]; diamond checks condition [1]; it directs to alternative Yes/No branches [1]; arrows show sequence/control flow [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**